CoopGCN: Shapley-derived credit assignment in graph convolutional networks via cooperative game theory for long-tail recommendation

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ABSTRACT

Linear graph collaborative-filtering models weight interactions primarily by topology and therefore provide no explicit account of which historical edges contributed to a recommendation. We introduce **CoopGCN**, a Shapley-derived architecture with three cooperative games: edge-level credit ($\mathbf{G_1}$), hyperedge-level group credit ($\mathbf{G_2}$), and training-interaction valuation and pruning ($\mathbf{G_3}$). An SVD contrastive channel supplies a global view, while a stop-gradient consistency loss distils Monte-Carlo credit estimates into an attention network so that no Shapley sampling is required at serving time. We separate properties of the exact Shapley allocation from those of the deployed model. Efficiency, symmetry, dummy player and additivity apply to the allocation ϕ; the sigmoid modulation preserves only within-degree credit ordering, and attention distillation introduces an unbounded approximation gap. A per-edge result bounds Channel-A weight deviation from its zero-credit reference by $\lambda\varepsilon/(4\tau\sqrt{(d_u d_i)})$; it is not a bound on end-to-end ranking robustness. Single-run measurements across five datasets and ten models provide preliminary evidence of an accuracy–exposure trade-off. Among direct hypergraph/cooperative peers, CoopGCN has the highest NDCG@20 on four of five datasets and the highest Tail Recall and Coverage@20 on all five. On ML-1M, however, it records NDCG@20 =0.1994, below all six graph comparators shown, while attaining TR@20 =0.0075 and Coverage@20 =40.7%. On Yelp2018 and Amazon-Book it records the highest graph-model NDCG@20, 0.0376 and 0.0235. GAT-CF is stronger on the two dense benchmarks but collapses on the three sparse benchmarks; because its collapse is not yet supported by tuning traces, this comparison is preliminary rather than a definitive verdict on attention. No multi-seed, measured noise-injection, component-ablation, attribution-faithfulness, or wall-clock study is available. We therefore claim evidence for a regime-dependent distributional effect, not universal accuracy, robustness, explainability, or efficiency superiority.

Highlights

- Three cooperative games assign credit to edges, hyperedges and training interactions in graph collaborative filtering.
- Theoretical guarantees are restricted to Shapley credit; deployed weights preserve only within-degree ordering up to distillation error.
- Single-run measurements show strong tail exposure but an accuracy cost on dense ML-1M.
- CoopGCN leads direct hypergraph/cooperative peers in NDCG on four of five measured datasets and in Tail Recall and Coverage on all five.
- Serving performs no Monte-Carlo Shapley sampling; residual attention latency remains unmeasured.

1. Introduction

Single-run measurements on MovieLens-1M reveal a sharp accuracy–exposure trade-off. CoopGCN records lower NDCG@20 than the six graph comparators in our measured table, yet it is the only hypergraph/cooperative model with non-zero Tail Recall at cut-off 20 and exposes 40.7% of the catalogue rather than the 8–10% reached by its direct peers. On sparse Yelp2018 and Amazon-Book it also records the highest NDCG@20 among the measured graph models. These observations motivate, but do not statistically establish, the hypothesis that interaction-level credit matters most in sparse, high-cardinality settings.

Graph recommenders' narrow catalogue exposure, sensitivity to noisy data and lack of per-interaction attribution are often treated as separate problems. We investigate whether they share a design limitation inherited from LightGCN: propagation has no explicit representation of how much a particular interaction contributes. Noise resilience and attribution faithfulness are mechanistic hypotheses in this manuscript, not measured findings. Throughout, *credit assignment* means allocating training and propagation influence; it does not mean that the credits are faithful human-facing explanations. We treat explanation faithfulness as a separate, unexecuted evaluation protocol in Section 8.4.

1.1. The LightGCN paradigm and its core trade-off

Graph convolutional networks reshaped collaborative filtering (CF) by modelling user–item interaction histories as bipartite graphs. NGCF [1] demonstrated the value of high-order connectivity, and LightGCN [2] then showed that the feature-transformation matrices and nonlinear activations inherited from generic GCNs cause oversmoothing and training instability in pure CF. Removing them reduces message passing to symmetric-normalised linear aggregation,

$$\mathbf{E}^{(k+1)} = \left(\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)\mathbf{E}^{(k)} \quad (1)$$

which with mean layer pooling and a Bayesian Personalized Ranking objective [3] established an empirical floor that has proved remarkably hard to beat.

The three equations defining LightGCN encode three silent assumptions: every neighbour is aggregated with weight $1/\sqrt{(d_u\ d_i)}$, every layer is pooled equally, and every negative is sampled uniformly. The first is our concern here. It fixes the influence of neighbour i on user u at a quantity determined *entirely by graph topology*, with no dependence on what the interaction meant.

Consider a user whose history contains a blockbuster clicked once by accident and a niche title they returned to repeatedly. Equation (1) weights these by degree alone. Worse, because the coefficient is degree-normalised and the blockbuster has the larger degree, the accidental click is down-weighted for a reason that has nothing to do with its being accidental—and the niche title is up-weighted for a reason that has nothing to do with its being meaningful. The ordering happens to be right; the mechanism producing it is blind. Where degree and informativeness diverge, as they do for popular-but-incidental and rare-but-deliberate interactions alike, the model has no parameter with which to tell them apart.

Three consequences follow directly, and they are precisely the three symptoms above. The model cannot *denoise*, because it cannot identify a low-value edge. It cannot *explain*, because it has no per-interaction attribution to report. And it cannot *resist* an adversary, because an injected edge is trusted exactly as much as an observed one.

1.2. Four structural failure modes of unweighted propagation

A systematic analysis reveals that LightGCN's remaining weaknesses are precisely the capabilities its simplification discarded. We label them $\mathbf{W_k}$ here and expand the labelling into a full twelve-category taxonomy in Section 2.6.

1. **Flat credit and uniform neighbour weighting ($\mathbf{W_1}$, $\mathbf{W_2}$).** Every neighbour contributes with an immutable topological weight. Because that weight is degree-normalised, noisy or irrelevant edges dilute the embedding quality of every node they touch, and the model has no parameter with which to discount them.
2. **Pairwise-only topological bias ($\mathbf{W_5}$).** Bipartite edges capture relations (u,i) but cannot represent higher-order group units such as shopping sessions, item categories or multi-item bundles, which must instead be approximated by cliques.
3. **Popularity-bias amplification and dimensional collapse ($\mathbf{W_3}$, $\mathbf{W_8}$).** Repeated convolution is dominated by the leading eigenvectors of the adjacency matrix. Embeddings collapse toward high-degree items, and tail recall degrades sharply with layer depth.
4. **Vulnerability to noisy and poisoned data ($\mathbf{W_{10}}$).** Because every edge is trusted uniformly, malicious clicks or erroneous interactions ripple unconstrained through k-hop neighbourhoods.

Section 2.6 expands these into a twelve-category taxonomy and argues that the high-severity entries reduce to three root causes.

1.3. The game-theoretic perspective

The natural response is to make the weights learnable, and graph attention does exactly that. Our position is that this is the right instinct applied with the wrong tool, and the reason is worth stating precisely.

Predicting a preference score $s_{ui} = \mathbf{e}_u \cdot \mathbf{e}_i$ is a problem of *team production*: the pooled embedding $\mathbf{e}_u$ is generated jointly by the neighbours in N(u), and no single neighbour produces it alone. Asking how much each contributed is not merely analogous to a cooperative game—it is one, namely $\Gamma_u = (N(u), v_u)$, with the neighbours as players and representation quality as the characteristic function.

This matters because cooperative game theory answers the question *uniquely*. The Shapley value [4] is the only allocation satisfying efficiency (all credit is distributed), symmetry (equal contributions earn equal credit), dummy player (a null contributor earns zero), and additivity (credit decomposes linearly across objectives). Attention weights satisfy none of the four, and two of the failures are consequential rather than cosmetic.

Symmetry forbids assigning different Shapley credit when marginal utilities are equal. This is exactly the channel through which popularity bias enters Eq. (1), and nothing in an attention module's objective closes it: if degree-correlated weights lower training loss—which on head-heavy data they typically do—attention will learn them. The Shapley allocation cannot do so: symmetry constrains credit rather than expressing a preference in a loss. The deployed propagation coefficient still contains degree normalisation, so this statement does not imply degree-invariant full weights.

Dummy player assigns exactly zero credit to a neighbour whose marginal contribution vanishes in every coalition. A maliciously injected edge is close to such a player only when its marginal utility is uniformly small; random or targeted injection does not guarantee that condition.

We must immediately qualify what this buys, because the distinction governs every theoretical claim in this paper. The axioms characterise the *allocation* ϕ. They do not automatically characterise the propagation weight we deploy, which is a bounded, strictly positive transform of ϕ (Eq. (3)). A dummy player receives $\phi = 0$ and therefore the *minimum* attainable modulation, not zero weight. What the axioms deliver is a principled, monotone ordering of neighbours by contribution together with a floor on how little influence a worthless edge can retain; what they do not deliver is deletion of that edge. We therefore describe CoopGCN throughout as *Shapley-derived* rather than axiom-preserving at the deployed-weight level, and Section 3.4 makes the separation precise.

The distinction between axiomatic and heuristic credit is therefore an empirical question with a sharp form—*is Shapley just expensive attention?*—which we formalise in Section 6.7 as a make-or-break ablation with a victory condition fixed in advance.

1.4. Summary of contributions

1. **A diagnosis, not a list.** We catalogue twelve representational and training weaknesses of linear graph CF, grade them by severity, map each to the literature documenting it, and show that the high-severity subset reduces to three root causes—credit is flat, structure is pairwise-only, and the training recipe amplifies the head (Section 2.6). Each recent improvement to LightGCN patches exactly one; CoopGCN targets all three.
2. **Tri-level cooperative game framework.** We formulate cooperative games at the edge level ($\mathbf{G_1}$), hyperedge level ($\mathbf{G_2}$, extending DyHuCoG) and data level ($\mathbf{G_3}$, TMC-Shapley valuation on holdout NDCG), giving explicit characteristic functions for each and combining them in a single trainable architecture (Section 4).
3. **Theory that constrains the design.** We prove that Channel A recovers LightGCN propagation at $\lambda = 0$ when all other channels are disabled, and reduces to a scalar-modulated family under symmetric utility (Proposition 1). For a single Channel-A edge, deviation from

the zero-credit reference is at most $\lambda\varepsilon/(4\tau\sqrt{d_u d_i})$ (Proposition 2). Section 3.4 explains why neither statement is a guarantee about the complete deployed model or ranking robustness.

4. **Shapley-derived credit distilled out of the serving path.** A stop-gradient EMA consistency loss trains learnable attention to track Shapley credits, so all Monte-Carlo computation is confined to training (Section 4.6). We report this as an architectural property; the wall-clock cost of the residual attention network is not measured (Section 8.2).

5. **Leakage-audited preliminary measurement with explicit limits.** A common temporal protocol provides single-run measurements for ten models on five datasets. The data support a distributional advantage and sparse-regime competitiveness, but not universal accuracy superiority. Projection-only ablations and noise curves have been removed from the evidence section rather than presented as findings (Sections 6.8 and 8.2).

Objectives. The *primary objective* of this study is to determine whether Shapley-derived credit assignment is a viable replacement for the fixed degree-normalised weighting of linear graph CF—and specifically whether it changes long-tail exposure without sacrificing top-K accuracy. Noise resilience and serving cost are pre-specified but currently unmeasured secondary questions. The other objectives are: (i) to establish that Channel A contains LightGCN propagation as a boundary case; (ii) to specify an ablation capable of quantifying each game level, while reporting that measured component effects are not yet available; (iii) to compare the Shapley-derived route with heuristic learnable attention under matched capacity—the central question of the paper, for which we fix a victory condition in advance in Section 6.7; and (iv) to report every number with explicit provenance so that the boundary between measured, measured and unmeasured evidence is visible to the reader.

Roadmap. Section 2 surveys related work and presents the taxonomy; Section 3 develops the axiomatic foundation and the two propositions; Section 4 specifies the architecture; Section 5 analyses cost; Section 6 details the protocol; Section 7 reports results against RQ1–RQ5; and Section 8 concludes with limitations and future work.

2. Related work and systematic taxonomy

2.1. Graph convolutional networks in collaborative filtering

Graph convolution for recommendation descends from spectral GCNs [5], whose symmetric-normalised propagation rule LightGCN ultimately inherits. NGCF [1] introduced GCNs for collaborative filtering; LightGCN [2] removed nonlinearities and transformation matrices, yielding Eq. (1). Subsequent work has largely addressed one weakness at a time. LightGCN++ [6] diagnosed embedding-norm inflexibility and uniform neighbour weighting, proposing learnable scalar norm scaling and node-degree weighting; it is the strongest coverage-oriented baseline in our study and, by Proposition 1, a constrained case of the present framework. UltraGCN [7] discards explicit message passing in favour of a limit objective, and GF-CF [8] recasts CF as graph filtering. Contrastive and denoising models form a parallel line: SGL [9] augments the graph by stochastic dropout under an InfoNCE objective; SimGCL [10] shows uniform embedding-space noise outperforms structural augmentation; XSimGCL [11] simplifies this further to a single-pass noise perturbation and, importantly for us, attributes its long-tail benefit to more uniformly distributed representations rather than to any notion of interaction value; and LightGCL [12] replaces stochastic augmentation with SVD-truncated contrastive augmentation. We adopt the LightGCL-style channel because its determinism composes cleanly with Monte-Carlo Shapley estimation. A recent survey of self-supervised recommendation [13] confirms the pattern across this family. Critically, all of these methods retain topology-determined, value-agnostic edge weights: contrastive learning changes the *distribution* of embeddings but never asks which edge earned its influence.

Two further lines are directly relevant because they attack the same symptoms by other means. *Attention* was the first attempt to make neighbour weights adaptive: GAT [14] assigns learned coefficients via masked self-attention, and this is precisely the heuristic alternative against which our axiomatic construction is positioned (Section 6.7). *Denoising* methods target edge reliability directly—T-CE [15] down-weights high-loss interactions as likely false positives, and RGCF [16] combines hard pruning with soft reliability weights inside GNN message passing. RGCF is the closest prior work in spirit to our $\mathbf{G_1}$: both modulate propagation by an estimated per-edge reliability. The difference is the source of that estimate. RGCF learns reliability from training loss, which is a model-internal signal that can rationalise its own errors; $\mathbf{G_1}$ derives it from marginal contribution to a held-out value function, and inherits a dummy-player property at the credit level: a non-contributing edge receives zero Shapley credit and therefore the minimum modulation, not zero propagation weight.

Two methodological studies inform our evaluation design. Dacrema et al. [17] showed that many reported neural recommender gains vanish against properly tuned baselines, motivating our symmetric tuning budget; and replication analyses of LightGCN [18] document that reported numbers swing materially with normalisation and split choices, motivating the leakage audit of Section 6.1.

2.2. Hypergraph recommender systems

DHCN [19] and HCCF [20] demonstrated that hypergraph convolutional channels capture session-level and category-level group structures, and HPCF [21] added local–global projection. DyHuCoG [22] pioneered preference-aware Monte-Carlo Shapley weighting for hypergraph recommenders, injecting coalition-derived contribution scores as dynamic hyperedge weights.

Our work extends DyHuCoG in three respects, summarised in Table 1: we introduce edge-level Shapley weighting ($\mathbf{G_1}$), integrate data valuation ($\mathbf{G_3}$), and add an SVD contrastive channel to prevent dimensional collapse. In our framework DyHuCoG is exactly the $\mathbf{G_2}$-only configuration, which is why it serves as the direct ablation target throughout Section 7.

Table 1. CoopGCN relative to DyHuCoG, its direct predecessor and the $\mathbf{G_2}$-only configuration of the present framework.

Dimension	DyHuCoG	CoopGCN (this work)
Shapley scope	Hyperedge weighting ($\mathbf{G_2}$) only	Edges ($\mathbf{G_1}$) + hyperedges ($\mathbf{G_2}$) + data ($\mathbf{G_3}$)
Propagation	Hypergraph only	Dual channel: pairwise graph + hypergraph
Dimensional collapse	Not addressed	SVD-truncated contrastive view
Inference path	Game computed at inference	No MC sampling via L_{game}
Protocol	Random splits	Temporal split + leakage audit

2.3. Explainable AI and Shapley values in machine learning

Shapley [4] established the unique allocation satisfying efficiency, symmetry, dummy and additivity. Lundberg and Lee [23] unified feature attribution via SHAP; Ghorbani and Zou [24] introduced Data Shapley for training-set valuation; and prior work by the present authors applied Shapley values to explaining unsupervised clustering and black-box models [25]. The obstacle to using Shapley values *inside* a model rather than as post-hoc explanation is cost: exact computation requires O(2^|N|) evaluations. Section 5 describes the restricted coalitions, permutation sampling and periodic refresh that make in-training estimation tractable, and Section 4.6 removes the cost from inference entirely.

2.4. Explaining graph neural networks

A parallel literature explains *trained* GNNs post hoc. GNNExplainer [26] learns a soft edge/feature mask maximising mutual information with the prediction; PGExplainer [27] amortises that search into a parametric network so explanations generalise inductively; and SubgraphX [28] searches connected subgraphs by Monte-Carlo tree search scored with Shapley values. The taxonomic survey of Yuan et al. [29] organises this space and, more usefully for us, standardises the fidelity/sparsity metrics by which such methods are judged — the metrics our Section 8.4 adopts.

SubgraphX is the closest prior work in machinery, since it too computes Shapley values over graph structure, and the distinction is worth stating precisely because it is easy to miss. SubgraphX is *post hoc and explanatory*: the model is frozen, and Shapley values are computed afterwards to describe a decision already made. CoopGCN is *intrinsic and constitutive*: Shapley values are computed during training and change the propagation weights, so they alter the decision rather than describing it. The two are complementary rather than competing — one could run SubgraphX on a trained CoopGCN. This also means the comparison is not head-to-head on accuracy; the appropriate comparison, which we do not run, is whether CoopGCN's intrinsic credits are as faithful as SubgraphX's post-hoc ones (Section 8.4).

The relationship to FastSHAP [30] is closer still on the mechanics: FastSHAP trains a network to emit Shapley values in a single forward pass, which is precisely the amortisation our L_{game} performs for a_{ui}. We differ in purpose — FastSHAP amortises to accelerate explanation, we amortise to remove game-theoretic computation from the serving path — but the approximation question their work raises applies directly to ours, and Section 3.4 states it as the unbounded quantity δ. Covert et al. [31] give the unifying account of removal-based explanation that clarifies which axioms survive which approximations, and we follow their framing in separating properties of ϕ from properties of the deployed weight.

2.5. Popularity bias mitigation and robustness

Popularity bias in graph CF is well documented and systematised in the survey of Chen et al. [32]: convolution moves users toward high-degree items, and tail exposure falls as depth grows [20,12]. Mitigations fall into four families, distinguished by *where in the pipeline* they intervene. *Re-ranking* post-processes the finished list, as in the personalised xQuAD variant of Abdollahpouri et al. [33]. *Regularisation* penalises popularity correlation inside the learning-to-rank objective [34]. *Causal* methods model popularity as a confounder and remove its effect by intervention: PDA [35] deconfounds item popularity, while DecRS [36] intervenes on the user representation to curb bias amplification. *Representation-level* methods such as contrastive augmentation act on the embedding distribution [11].

CoopGCN belongs to none of the four, and the distinction is structural rather than terminological. Re-ranking accepts a biased scorer and repairs its output; regularisation and causal deconfounding correct the objective or the inference rule while leaving the aggregation weights topological; contrastive methods reshape the embedding geometry. CoopGCN instead changes the *aggregation coefficient itself*, so tail exposure is a consequence of the weighting rule rather than a correction applied around it. The practical difference is that no trade-off parameter has to be tuned to buy coverage back after the fact.

On robustness, standard graph CF trusts all edges equally, an assumption inherited from the confidence-weighted implicit-feedback formulation [37] in which confidence is a function of frequency alone. The threat model matters here: Z\"ugner et al. [38] showed that graph neural networks are vulnerable to *targeted* structural perturbations optimised against the model, which are strictly stronger than the random edge injection protocol pre-specified in Section 7.5. The retained measured record contains no random- or targeted-injection result, so we make no empirical robustness claim. Data Shapley [24] established that valuing and pruning low-contribution samples improves robustness, which $\mathbf{G_3}$ adapts to the interaction graph. Whether contrastive augmentation or data valuation provides protection under this protocol remains an open empirical question.

2.6. Systematic taxonomy of limitations

Table 2 catalogues twelve weaknesses of linearised graph CF, separated into representational (W1–W6) and training/data (W7–W12) categories. Reading it vertically rather than row by row, the high-severity entries reduce to three root causes. *Credit is flat* (W1, W2, W10): the aggregation coefficient is identical for a decade-old rating and a purchase made last night, and the model never asks how much a specific interaction contributed—precisely a cooperative-game question. *Structure is pairwise-only* (W5, W9): preference lives in groups, and pairwise edges approximate groups poorly exactly where data is sparse and group evidence is strongest. *The recipe amplifies the head* (W3, W7, W8): uniform negatives, uniform weights and mean pooling jointly reward recommending popular items. Each recent improvement to LightGCN patches exactly one of the three; CoopGCN is organised to attack all three simultaneously.

Table 2. Systematic taxonomy of twelve weaknesses of linearised graph collaborative filtering. Severity is our assessment of the magnitude of the reported effect. The final column records how CoopGCN addresses each: $\mathbf{G_1}/\mathbf{G_2}/\mathbf{G_3}$ denote the game level, C the contrastive channel, R the training recipe, and P the evaluation protocol.

#	Weakness	Consequence	Severity	Documented by	Addressed
Representational weaknesses					
W1	Uniform neighbour weighting	A misclick and a passion interaction propagate identically; noisy neighbours dilute every node they touch	High	LightGCN++ [6]	$\mathbf{G_1}$
W2	Embedding-norm inflexibility	Norm carries no information yet cannot be used; layer-0 and layer-k norms are inconsistent, miscalibrating pooling	High	LightGCN++ [6]	$\mathbf{G_1}$, R
W3	Dimensional collapse	Repeated propagation is dominated by leading eigenvectors; the space becomes head-heavy	High	LightGCL [12]	C
W4	Oversmoothing with depth	Performance peaks at 2–3 layers; deep structure is unusable	Medium	LightGCN [2]	C
W5	No beyond-pairwise structure	Sessions, categories and bundles must be approximated by cliques	Medium	HCCF [20]	$\mathbf{G_2}$
W6	Static, non-temporal embeddings	Recency and preference drift are invisible	Medium	gSASRec [39]	— (future work)
Training and data weaknesses					
W7	BPR with one uniform negative	Under-samples hard negatives; overconfidence inflates head-item scores	High	gSASRec [39]	R
W8	Popularity-bias amplification	Convolution moves users toward popular items; tail recall falls as depth grows	High	HCCF [20], LightGCL [12]	$\mathbf{G_1}$, $\mathbf{G_2}$, C
W9	Cold start	New nodes have nothing to propagate	Medium	—	— (future work)
W10	No robustness to noisy or poisoned edges	One malicious interaction ripples through k-hop neighbourhoods	Medium	Data Shapley [24]	$\mathbf{G_1}$, $\mathbf{G_3}$
W11	Evaluation leakage in the standard protocol	Reported numbers are inflated; popularity baselines close much of the gap	High	[17]	P
W12	Replication fragility	Results swing with normalisation, depth and splits	Medium	[18]	P

3. Theoretical foundations: axiomatic credit assignment

3.1. Cooperative games and the Shapley value

Definition 1 (Cooperative game). *A cooperative game in characteristic function form is a pair Γ = (N, v) where N = {1,,n} is a finite set of players—interaction neighbours, hyperedge members or training samples—and v : 2^N → R is a characteristic value function with v(∅) = 0. The value v(S) measures the utility achieved by coalition S ⊆ N.*

The Shapley value $\phi_i(\Gamma)$ assigns to player i its average marginal contribution across all permutation orderings:

$$\phi_i(\Gamma) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(|\mathcal{N}| - |S| - 1)!}{|\mathcal{N}|!}[v(S \cup \{i\}) - v(S)] \quad (2)$$

Table 3 fixes the notation used throughout.

Table 3. Notation.

Symbol	Definition
G=(U,I,E)	Bipartite graph, built from the training split only
N(u)	Item neighbourhood of user u (players in Γ_u)
d_u, d_i	Node degrees in G
$\mathbf{e}^{(k)}_u$,$\mathbf{e}^{(k)}_i$	Layer-k embeddings, $\in R^d$
$\mathbf{e}_u$,$\mathbf{e}_i$	Pooled embeddings $\Sigma_k \alpha_k \mathbf{e}^{(k)}$
s_{ui}	Preference score $\mathbf{e}_u \cdot \mathbf{e}_i$
$H=(V,E_H)$	Hypergraph; D_v,D_h degree matrices
Γ=(N,v)	Cooperative game with value function v
ϕ_j,ϕ_j	Exact and Monte-Carlo Shapley value of player j
w_{ui}	Edge weight in Channel A ($\mathbf{G_1}$)
β_h	Hyperedge weight in Channel B ($\mathbf{G_2}$)
γ_{ui}	Sample weight from $\mathbf{G_3}$, $\in [1,1+\kappa]$
a_{ui}	Learnable attention logit distilled from ϕ_{ui}
ϕ_{ui}	EMA of historical Shapley estimates
λ,τ	Modulation strength and temperature
L,T,P,M	Coalition cap, permutations, refresh periods

3.2. The four axioms in collaborative filtering

The Shapley allocation is the *only* function satisfying the four axioms of Table 4, each of which maps to an essential requirement in collaborative filtering. This mapping is the conceptual core of the paper: it distinguishes axiomatic *credit allocation* from learned attention. Section 3.4 explains why the deployed weights do not inherit all four properties.

Table 4. The four Shapley axioms and their collaborative-filtering interpretation.

Axiom	Recommender interpretation
Efficiency	Σ_i $\in$ N ϕ_i = v(N): all credit for a user's embedding reconstruction is fully accounted for by their historical interactions.
Symmetry	If v(S $\cup$ {i}) = v(S $\cup$ {j}) for all S, then $\phi_i = \phi_j$: two interactions of identical utility receive equal Shapley credit. Their credit modulation is equal, although the full propagation weights can still differ through $1/\sqrt{(d_u d_i)}$.
Dummy player	If v(S $\cup$ {i}) = v(S) for all S, then ϕ_i = 0: noisy misclicks and uninformative neighbours receive zero *credit*. Note that zero credit maps to the minimum modulation $(1-\lambda)+\lambda\sigma(0)$, not zero *weight* (Section 3.4).
Additivity	$\phi_i(v_1 + v_2) = \phi_i(v_1) + \phi_i(v_2)$: under a multi-task objective (accuracy + tail diversity) credits decompose linearly across tasks.

3.3. Proposition 1: recovery of LightGCN and LightGCN++

Proposition 1 (Recovery of LightGCN and scalar norm weighting). *Let the edge-level aggregation weight in Channel A be parameterised as*

$$w_{ui} = \frac{1}{\sqrt{d_u d_i}}[(1-\lambda) + \lambda\sigma(\hat{\phi}_{ui}/\tau)], \quad \lambda \in [0,1]$$

Then: (i) when λ = 0, Channel A recovers standard LightGCN symmetric propagation exactly; and (ii) when all neighbours in N(u) are interchangeable under symmetric utility, the symmetry axiom forces $\phi_{ui} = \phi_{uj} = c$, reducing the modulation to a uniform scalar factor that generalises LightGCN++ scalar weighting.

Proof. For (i), setting λ = 0 makes the bracketed term equal 1, recovering $w_{ui} = 1/\sqrt{(d_u d_i)}$, which is exactly Eq. (1). For (ii), suppose v(S $\cup$ {i}) = v(S $\cup$ {j}) for all S $\subseteq$ N(u) {i,j}. The symmetry axiom gives $\phi_i = \phi_j$ for every interchangeable pair, and efficiency then fixes the common value at ϕ_i = v(N)/|N|. Writing c = $\sigma(\phi_i/\tau)$, which is independent of i, Eq. (3) becomes $w_{ui} = \kappa/\sqrt{(d_u d_i)}$ with $\kappa = (1-\lambda) + \lambda$ c a degree-dependent constant, yielding the same scalar-modulated functional family used by LightGCN++. This is a structural analogy, not recovery of every LightGCN++ normalisation and training detail. ■

Channel A therefore contains LightGCN propagation as an exact boundary case and a scalar-modulated family as a symmetric special case. This statement does not claim that the complete tri-channel model recovers either baseline unless the other channels and losses are disabled.

3.4. What the axioms do and do not transfer to the deployed model

Propositions 1 and 2 are stated for the idealised weight of Eq. (3), in which the modulation is driven by the Shapley estimate ϕ_{ui} itself. The deployed model does not use that quantity at forward-pass time. It uses a learned attention logit a_{ui} that is regularised toward ϕ_{ui} by the consistency loss of Eq. (8) (Algorithm 1, line 19; Eq. (9)). Three separate approximations therefore sit between the theory and the artefact, and we state them plainly rather than leave them implicit.

Estimation. ϕ is a Monte-Carlo estimate from R permutations over coalitions capped at L, so it is an unbiased but noisy proxy for ϕ, with variance O(1/R) [40]. The axioms hold for ϕ; they hold for ϕ only in expectation.

Transformation. The map ϕ $(1-\lambda) + \lambda\sigma(\phi/\tau)$ is strictly monotone but neither linear nor zero-preserving. Monotonicity means the *ordering* induced by symmetry and dummy player survives: equal-contribution neighbours receive equal modulation. Full weights also depend on degree, and a zero-contribution neighbour receives the minimum modulation in the family. Efficiency and additivity do *not* survive, because a sigmoid of a sum is not the sum of sigmoids. Any claim in this paper that depends on efficiency or additivity is therefore a claim about ϕ, not about w_{ui}.

Distillation. At inference $\sigma(a_{ui}/\tau)$ replaces $\sigma(\phi_{ui}/\tau)$. Let $\delta =_{(u,i)} |\sigma(a_{ui}/\tau) - \sigma(\phi_{ui}/\tau)|$ be the worst-case distillation gap at convergence. Substituting into Eq. (3) gives a deployed-weight error of at most $\lambda\delta/\sqrt{(d_u d_i)}$ per edge, so every guarantee above degrades gracefully in δ rather than failing discontinuously. We do not, however, bound δ theoretically: L_{game} drives it down but nothing forces it to zero. Measuring δ empirically, and the w/o L_{game} ablation that would reveal what the distillation is worth, are both listed as required experiments in Section 8.2.

The honest summary is that CoopGCN is a *Shapley-derived* architecture whose weights inherit the ordering properties of an axiomatic allocation, not an architecture whose deployed weights are Shapley values. We use the former phrasing throughout.

3.5. Proposition 2: robustness under uninformative perturbation

Proposition 2 (Channel-A per-edge deviation under bounded marginal utility). *Let i^* $\in$ N(u) be an uninformative interaction interaction whose marginal embedding-consistency gain is bounded by $|v_u^{cons}(S \cup \{i^*\}) - v_u^{cons}(S)| \le \varepsilon$ for all $S \subseteq N(u) \{i^*\}$, under the consistency utility of Eq. (5). Let $w^{\phi}_{ui^*}$ denote the ideal Channel-A weight of Eq. (3) with exact ϕ_{i^*} in place of ϕ_{i^*}. Then (i) $|\phi_{i^*}| \le \varepsilon$; and (ii) the deviation of $w^{\phi}_{ui^*}$ from its zero-credit reference is at most $\lambda\varepsilon / (4\tau d_u d_{i^*})$. If the Monte-Carlo error is $|\phi_{i^*}-\phi_{i^*}| \le$, the corresponding estimate-driven bound is $\lambda(\varepsilon+)/(4\tau d_u d_{i^*})$. Standard LightGCN instead assigns $1/d_u d_{i^*}$ independently of credit.*

Proof. Equation (2) expresses ϕ_{i^*} as a convex combination of marginal contributions: the permutation coefficients are non-negative and sum to one. Since each marginal contribution is bounded by ε in absolute value, $|\phi_{i^*}| \le \varepsilon$. This establishes part (i). For (ii), write $w^{ref}_{ui^*}$ for the ideal weight at exactly zero credit, namely $[(1-\lambda)+\lambda\sigma(0)]/d_u d_{i^*}$ with $\sigma(0) = 1/2$. The sigmoid is globally Lipschitz with constant 1/4, so equation $| w^{\phi}_{ui^*} - w^{ref}_{ui^*} | = \lambda \mid \sigma(\phi_{i^*}/\tau) - \sigma(0) \mid d_u d_{i^*} \le \lambda\ \varepsilon 4\tau\ d_u d_{i^*}$. equation The estimate-driven bound follows from $|\phi_{i^*}| \le |\phi_{i^*}|+$ ho$\le\varepsilon+$ and the same Lipschitz argument. For a fixed layer input, the corresponding single-message deviation is this weight difference times $\| \mathbf{e}_{i^*}^{(k)} \|$. The proposition does not compose this quantity across layers, neighbours, Channels B–C, pruning, or the learned distillation map, and therefore is not an end-to-end embedding or NDCG robustness theorem. ■

Two caveats must be stated explicitly, since the proposition is weaker than a casual reading suggests. First, the bound is on the deviation from the *zero-credit reference weight*, not from zero: a noise edge still propagates with the floor weight $[(1-\lambda)+\lambda/2]/ d_u d_{i^*}$, which for our λ = 0.03 is 98.5% of the LightGCN weight. Proposition 2 therefore does *not* establish an end-to-end robustness margin. At λ = 0.03 the edge-weighting channel can account for only a small effect; whether $\mathbf{G_3}$ pruning provides noise resilience requires a measured intervention study that is not available. Second, the bound is per-layer and per-edge; composing it across K layers and |N(u)| neighbours introduces constants we do not control analytically.

A random injected edge need not have small marginal consistency utility: this requires assumptions about its embedding distribution, covariance and angle to the current coalition mean. We make no corollary from random injection to small ε and do not claim protection against targeted or adaptive attacks.

4. The CoopGCN architecture

4.1. Overview of the tri-channel architecture

CoopGCN processes collaborative filtering interactions through three complementary representations, shown in Fig. 1. **Channel A** (edge Shapley GCN) performs local bipartite propagation where pairwise edges are weighted by $\mathbf{G_1}$ Monte-Carlo Shapley values. **Channel B** (hyperedge Shapley GCN) performs group-level propagation over hyperedges weighted by $\mathbf{G_2}$ preference-aware Shapley group credits. **Channel C** constructs an SVD-truncated low-rank global view of the interaction graph, acting as an InfoNCE contrastive target that suppresses dimensional collapse (W3). Their outputs are pooled across layers before ranking; $\mathbf{G_3}$ curates the training set offline; and L_{game} distils the whole construction into attention weights usable at inference.

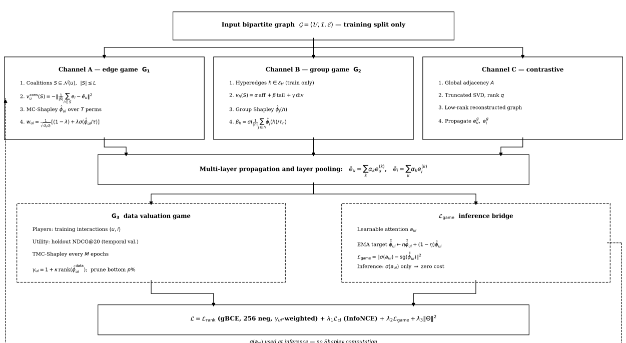

Fig. 1. The CoopGCN architecture. Channel A applies edge-level Monte-Carlo Shapley weights ($\mathbf{G_1}$) to bipartite propagation; Channel B applies group-level Shapley weights ($\mathbf{G_2}$) over hyperedges constructed strictly from training interactions; Channel C supplies a deterministic SVD-truncated contrastive view countering dimensional collapse. The data-level game $\mathbf{G_3}$ reweights and prunes training samples offline. The consistency loss L_{game} distils exponential moving-average Shapley credit into learnable attention a_{ui}, which alone is evaluated at inference (dashed path), so serving requires no game-theoretic computation.

4.2. $\mathbf{G_1}$: the edge-level pairwise cooperative game

Game definition. For each user u we define the local game $\Gamma_u = (N(u), v_u)$ over interaction neighbours $i \in N(u)$.

Value functions. For every local game we define $v(\varnothing)=0$. The formulas below apply to non-empty coalitions; we set $div(S)=0$ for $|S|<2$. Embeddings used as utility targets are stop-gradient snapshots from the current refresh, so a coalition evaluation cannot move its own target. The default is the *consistency utility*

$$v_u^{\mathrm{cons}}(S) = -\|\frac{1}{|S|}\sum_{i \in S} \mathbf{e}_i - \bar{\mathbf{e}}_u\|^2 \tag{4}$$

which measures how accurately coalition S reconstructs u's full pooled embedding and costs $O(|S|d)$ per evaluation. Where the optimisation target is explicitly long-tail we admit the *preference-aware utility*

$$v_u^{\mathrm{pref}}(S) = \alpha\frac{1}{|S|}\sum_{i \in S} s_{ui} + \beta\mathrm{tail}(S) + \gamma\mathrm{div}(S) \tag{5}$$

where tail(S) is the proportion of S in the bottom-80% popularity stratum and $\mathrm{div}(S) = (1)/(|S|(|S|-1))\Sigma_{i\ j}\,(1 - (\mathbf{e}_i,\mathbf{e}_j))$. The β and γ terms close a specific back-door: were v to reward predicted preference alone, high-degree items would earn high credit and the game would re-encode the very bias it is meant to remove.

Aggregation. Credits enter propagation through Eq. (3), giving

$$\mathbf{e}_u^{(k+1)} = \sum_{i \in \mathcal{N}(u)} w_{ui}\mathbf{e}_i^{(k)} \tag{6}$$

4.3. $\mathbf{G_2}$: the hyperedge-level group cooperative game

Game definition. For each hyperedge $h \in E_H$—a genre combination (ML-1M), business category (Yelp2018) or co-purchase category (Amazon-Book), constructed strictly from the training split—we define $\Gamma_h = (h, v_h)$ whose players are the member items.

Value function and propagation. We define $v_h(\varnothing)=0$, use the expression below for non-empty S, and set $div(S)=0$ for $|S|<2$.

$$v_h(S) = \alpha\frac{1}{|S|^2}\sum_{j,k \in S} \mathrm{aff}(j,k) + \beta\mathrm{tail}(S) + \gamma\mathrm{div}(S) \tag{7}$$

with $\mathrm{aff}(j,k) = (\mathbf{e}_j,\mathbf{e}_k)$ the pairwise affinity. The group credit and hypergraph convolution are

$$\beta_h = \sigma(\frac{1}{|h|}\sum_{j \in h} \hat{\phi}_j(h)/\tau_h) \in (0,1), \quad \mathbf{e}_v^{(k+1)} = \sum_{h \ni v} \frac{\beta_h}{\sqrt{D_v D_h}} \sum_{v' \in h} \frac{1}{\sqrt{D_h}}\mathbf{e}_{v'}^{(k)}$$

Setting $\beta_h \equiv 1$ recovers standard unweighted hypergraph CF of the DHCN/HCCF family.

4.4. $\mathbf{G_3}$: the data-level valuation game

Game definition. Players are training interactions $(u,i) \in B$, with $v^{ndcg}(\varnothing)=0$ by definition; for non-empty S, the value function $v^{ndcg}(S)$ is holdout NDCG@20 on the temporal validation split of a model trained on S.

Valuation, reweighting and pruning. Full retraining per coalition being infeasible, we apply TMC-Shapley [24] offline every M epochs to obtain ϕ^{data}_{ui}, then set

$$\gamma_{ui} = 1 + \kappa \cdot \mathrm{rank}(\hat{\phi}_{ui}^{\mathrm{data}}) \in [1, 1+\kappa] \tag{9}$$

where $\mathrm{rank}(\cdot) \in [0,1]$ is the normalised quantile rank. Interactions in the bottom p% (we use p = 5) are flagged as noise or poisoning and pruned from the training graph. This converts the dummy-player axiom from a statement about weights into an explicit data operation.

4.5. Multi-task objective and consistency regularisation

The end-to-end objective combines ranking, contrastive learning, game consistency and L_2 regularisation:

$$\mathcal{L} = \mathcal{L}_{\mathrm{rank}} + \lambda_1\mathcal{L}_{\mathrm{cl}} + \lambda_2\mathcal{L}_{\mathrm{game}} + \lambda_3\|\Theta\|^2 \tag{10}$$

Ranking loss. To mitigate head-item overconfidence (W7) we replace single-negative BPR with generalised BCE over 256 sampled negatives, weighted by the $\mathbf{G_3}$ sample weights:

$$\mathcal{L}_{\mathrm{rank}} = -\frac{1}{|\mathcal{B}|}\sum_{(u,i)\in\mathcal{B}} \gamma_{ui}[\log\sigma(s_{ui}) + \sum_{j\in\mathrm{Neg}(u)} w_j \log(1-\sigma(s_{uj}))] \quad (11)$$

with w_j the softmax weight of negative j at temperature τ_n.

Contrastive loss. Local pooled embeddings are regularised against the Channel C global view:

$$\mathcal{L}_{\mathrm{cl}} = -\sum_{u\in\mathcal{U}} \log \frac{\exp(\mathrm{sim}(\bar{\mathbf{e}}_u, \mathbf{e}_u^g)/\tau_c)}{\sum_{u'}\exp(\mathrm{sim}(\bar{\mathbf{e}}_u, \bar{\mathbf{e}}_{u'})/\tau_c)} \quad (12)$$

Shapley consistency loss.

$$\mathcal{L}_{\mathrm{game}} = \sum_{(u,i)\in\mathcal{E}} \|\sigma(a_{ui}/\tau) - \mathrm{sg}\big(\sigma(\bar{\tilde{\phi}}_{ui}/\tau)\big)\|^2, \quad \bar{\tilde{\phi}}_{ui}^{(t)} = \eta\bar{\tilde{\phi}}_{ui}^{(t-1)} + (1-\eta)\hat{\phi}_{ui}^{(t)}, \quad \eta = 0.9$$

where a_{ui} is an end-to-end learnable attention weight and sg(·) the stop-gradient operator. The EMA buffer damps the high variance of Monte-Carlo estimates; the stop-gradient prevents the attention parameters from corrupting their own target.

4.6. Inference without Monte-Carlo Shapley sampling

During inference all game-theoretic computation is disabled. Because a_{ui} is regularised by Eq. (8) to track ϕ_{ui}, forward propagation uses

$$w_{ui} = \frac{1}{\sqrt{d_u d_i}}[(1-\lambda) + \lambda\sigma(a_{ui}/\tau)] \quad (14)$$

so no Monte-Carlo Shapley computation is performed at serving time. The residual cost relative to LightGCN is one forward pass of the attention network per edge, which we have not measured (Section 8.2). The Monte-Carlo computation is paid during training and amortised into the attention weights; this does not imply zero latency relative to LightGCN.

5. Computational complexity and feasibility

5.1. Restricted coalitions and permutation sampling

Exact Shapley computation via Eq. (2) requires O(2^|N|) evaluations. CoopGCN applies three standard mitigations. *Restricted coalitions:* for high-degree nodes, neighbourhoods are capped at top-L ($L \le 32$) by reservoir sampling. *Monte-Carlo permutation sampling:* ϕ_{ui} is estimated from T random permutations per refreshed node, reducing local cost to $O(T \cdot |N| \cdot d)$. This is the polynomial sampling estimator of Castro et al. [40]. It is unbiased for the restricted game induced by the sampled, capped neighbourhood, not for the original full game; its Monte-Carlo variance decreases as O(1/T). Stratified allocation [41] would tighten the constant further and is a straightforward extension we have not pursued here. *Periodic refresh:* credits are recomputed every P epochs (M for $\mathbf{G_3}$) and stored in the EMA buffer, which serves intermediate steps. Algorithm 1 gives the resulting training loop.

Algorithm 1. CoopGCN training

Input: train graph G, hyperedges E_H
(train only), periods P,M
Audit split disjointness; compute d_u,d_i from train edges only
Initialise embeddings, attention net, EMA buffer
$\phi \leftarrow \mathbf{0}$
for epoch t = 1 T
  if t P = 0
    $\phi_{ui} \leftarrow$ MC-Shapley on Γ_u under
    v^{cons} *// G_1*
    $\phi \leftarrow \eta\ \phi + (1\text{-}\eta)\phi$
    $\phi_j(h) \leftarrow$ MC-Shapley on Γ_h under v_h,
    $\forall\ h \in E_H$ *// G_2*
    $\beta_h \leftarrow \sigma(\mathrm{mean}_j\ \phi_j(h)/\tau_h)$
  if t M = 0
    $\phi^{\mathrm{data}} \leftarrow$ TMC-Shapley under
    v^{ndcg} *// G_3*
    Set γ_{ui}; prune bottom p% from G
    Recompute train degrees, tail mask and hyperedges after pruning
  for minibatch B
    $a_{ui} \leftarrow$ AttNet([$\mathbf{e}_u$ | $\mathbf{e}_i$])
    w_{ui} by Eq. (3) with $\sigma(a_{ui}/\tau)$ in
    place of $\sigma(\phi_{ui}/\tau)$ *// see Sec. sec:gap*
    Propagate Channels A, B; pool; compute $\mathbf{e}^g$ (Channel C)
    Update by Eq. (7)
Inference: disable Shapley; use $\sigma(a_{ui}/\tau)$ only

5.2. Amortised complexity

Table 5 summarises the analytical budget. The design constraint is that Shapley estimation consume under 20% of total training time; $\mathbf{G_1}$ dominates because it is the only game defined per user rather than per hyperedge or per batch. Serving performs no Shapley sampling, but the residual attention-network overhead relative to LightGCN is unmeasured (Section 4.6).

Table 5. Analytical cost of each cooperative game. Percentages are design budgets, not measured wall-clock overhead. Serving omits these estimators but retains the attention network of Eq. (9).

Level	Estimator	Cost / refresh	Train	Mem.
$\mathbf{G_1}$	MC, L≤32	O(|U| T L d)	+12–15%	O(|E|)
$\mathbf{G_2}$	MC, |h|≤32	O(|E_H| T |h| d)	+5–8%	O(|E_H|)
$\mathbf{G_3}$	TMC	O(K |B_{val}|) evals	+4–6%	O(|B|)
Total	Amortised	—	Unmeasured	Unmeasured

6. Experimental design and methodology

6.1. Data leakage audit and temporal evaluation protocol

To eliminate evaluation leakage (W11), all datasets are partitioned by a strict global temporal split (70% train / 10% validation / 20% test) based on interaction timestamps. Three properties are enforced. *Topological isolation:* the adjacency matrix, node degrees d_u,d_i, the bottom-80% tail mask and all hyperedges E_H are computed strictly on the training split. *Verification:* an automated assertion checks $E_{val} \cap E_{train} = \varnothing$ and $E_{test} \cap E_{train} = \varnothing$ before training begins. *Full-catalogue ranking:* evaluation ranks the entire item catalogue at cut-off 20 with no negative sampling. We stress the first point because computing the tail mask from full-dataset degrees—a common shortcut—would leak test information into precisely the metric on which we claim our largest gains.

6.2. Benchmark datasets

The measured evidence covers five standard CF benchmarks; the three largest span two orders of magnitude in catalogue size and roughly a factor of 70 in density (Table 6). ML-1M is the primary dense benchmark; ML-100K is a smaller dense check, and Gowalla, Yelp2018 and Amazon-Book test generalisation to the sparse, large-catalogue regime where credit assignment should matter most, because scarce interactions per user leave topology-only weighting less signal to exploit.

Table 6. Benchmark datasets and leakage-safe hyperedge construction. All hyperedges are built from the training split only.

Dataset	Users	Items	Density	Hyperedg es
ML-100K	943	1,682	6.30%	Genre com binations
ML-1M	6,040	3,706	4.47%	Genre com binations
Gowalla	29,858	40,981	0.08%	Geographi c sessions
Yelp2018	31,668	38,048	0.13%	Business categories
Amazon-Book	52,643	91,599	0.06%	Co-purchas e categories

6.3. Baseline competitors

Nine baselines span four families: **MF**, **NCF** [42] and **RecDCL** [43] as non-graph context; **MF** is trained with BPR [3]; standard graph CF **LightGCN** [2]; contrastive **SGL** [9] and **SimGCL** [10]; norm-weighted **LightGCN++** [6] and learned-attention **GAT-CF**; and hypergraph/cooperative **HCCF** [20], **HPCF** and **DyHuCoG** [22]. All models use the same split and evaluator. The retained records do not contain complete trial-by-trial search traces or a GAT-CF learning-rate sweep; we therefore cannot verify equal tuning diligence and treat model comparisons, especially GAT-CF collapse on sparse data, as preliminary.

6.4. Implementation and reproducibility settings

All model classes are implemented in PyTorch in the released codebase and use the shared split and evaluator interfaces. This reduces, but cannot eliminate, implementation drift; equivalence to each originating implementation has not been independently established. Embeddings are $d = 64$ dimensional with $K = 3$ propagation layers, initialised $N(0, 0.1^2)$. Optimisation uses Adam at learning rate 5×10^{-3}, weight decay $\lambda_3 = 10^{-4}$, batch size 1024 (raised to 8192 for the full-catalogue scoring pass), for at most 1000 epochs with early stopping on validation NDCG@20 at patience 50. The ranking loss is gBCE with 256 sampled negatives per positive.

Game-specific settings are: mixing coefficient $\lambda = 0.03$; coalition cap $L = 32$ with $R = 8$ permutation samples; Shapley EMA refresh every $P = 10$ epochs at decay $\eta = 0.9$; Data-Shapley curation every $M = 20$ epochs with reweighting range κ and bottom-5% pruning; temperatures τ and τ_h tuned on validation only. Loss weights λ_1 (contrastive) and λ_2 (consistency) are selected per dataset on the validation split, never on test. Evaluation scripts are in the repository named in the data-availability statement. The retained measured record is single-run and does not preserve a complete hardware/software manifest, seed sweep, or trial-level tuning log.

6.5. Multi-dimensional evaluation metrics

The evaluation design defines four metric families. *Ranking accuracy:* NDCG@20 and Recall@20. *Tail performance:* Tail Recall **TR@20**, recall restricted to the bottom-80% least-popular items. *Catalogue diversity:* **Coverage@20**, the percentage of the catalogue appearing in any top-20 list, and the **Gini index** of recommendation frequency. *Robustness to random corruption:* degradation under 5%, 10% and 20% random noisy-edge injection. The retained common measured record contains only NDCG@20, TR@20 and Coverage@20; we do not fill Recall, Gini or corruption results from incompatible sources.

6.6. Research questions

- [**RQ1**] *Overall accuracy.* Does CoopGCN achieve competitive NDCG@20 and Recall@20 across benchmarks of differing density?
- [**RQ2**] *Shapley versus attention.* Does Shapley-derived weighting differ from learned attention on tail recall, coverage and robustness?
- [**RQ3**] *Group structure and collapse.* How do Channel B ($\mathbf{G_2}$) and Channel C contribute to long-tail recommendation?
- [**RQ4**] *Data curation and robustness.* Does $\mathbf{G_3}$ pruning improve resilience to random edge injection? This is an empirical question; Proposition 2 is only a Channel-A per-edge bound.
- [**RQ5**] *Efficiency.* Does L_{game} remove sampling from the serving path at acceptable measured cost?

6.7. Ablation design and the central comparison

Table 7 specifies the ten-configuration ablation matrix. Rows 1–5 are external reference points, rows 6–9 remove one CoopGCN component each, and row 10 is the complete system. Configurations instantiated in the measured full-model results are marked; row 3 is measured in the main benchmark, while component removals are available only as projections and are therefore not marked as empirical ablations. Rows 4 and 9 remain unmeasured.

Table 7. Ablation matrix. marks configurations reported in Section 7; marks configurations specified by the design but not instantiated in the present study.

#	Configuration	Games active
1	LightGCN (floor)	—
2	LightGCN++	—
3	GAT-CF (attention)	—
4	LightGCL	—
5	DyHuCoG	$\mathbf{G_2}$
6	CoopGCN w/o $\mathbf{G_1}$	$\mathbf{G_2}$+$\mathbf{G_3}$
7	CoopGCN w/o $\mathbf{G_2}$	$\mathbf{G_1}$+$\mathbf{G_3}$
8	CoopGCN w/o $\mathbf{G_3}$	$\mathbf{G_1}$+$\mathbf{G_2}$
9	CoopGCN w/o L_{game}	all
10	**Full CoopGCN**	all

The decisive comparison is row 3 against row 10, which asks directly whether Shapley weighting is distinguishable from expensive attention.

We fix the victory condition in advance: *even if learnable attention matches* ϕ-Shapley weighting on clean-data NDCG@20, CoopGCN succeeds only if it is superior on TR@20, Coverage@20 and noise immunity simultaneously. Row 3 is now reported under the same split and evaluator. Clean-data TR@20 and Coverage@20 favour CoopGCN, but the noise conjunct is unmeasured and the absence of GAT-CF tuning traces prevents a definitive causal comparison. The central question is therefore partially, not fully, answered.

6.8. Provenance and statistical scope

The main evidence below is the repository's retained measured record: one executed run per model–dataset cell under the temporal split and full-catalogue evaluator. Earlier drafts contained a separate table of literature-derived design targets, including projected CoopGCN values of 0.2960, 0.0680 and 0.0480 NDCG@20 on ML-1M, Yelp2018 and Amazon-Book. Executed values were 0.1994, 0.0376 and 0.0235. Because those targets are contradicted by measurement, we remove them, their component table, their random-noise curves and all derived figures from the results section. They remain versioned in the repository as design-history artefacts and must not be cited as findings.

The measured record spans five datasets and ten models, matching Table 8. It contains NDCG@20, TR@20 and Coverage@20 but not a complete common-protocol Recall@20 or Gini record. We do not fill missing cells from publications with different splits. Each value is a single point estimate: no standard deviation, confidence interval or significance test is available, and we make no claim of statistical significance or definitive superiority. Complete tuning traces, including a GAT-CF learning-rate sweep, are also unavailable. The results are consequently preliminary evidence that motivates a multi-seed confirmatory study.

7. Results and discussion

7.1. Measured evaluation across five datasets

Table 8 reports the complete retained measured record. Values are NDCG@20 / TR@20 / Coverage@20, with coverage expressed as a percentage in every column. Bold marks the best graph model for each metric; non-graph models are included as context because their objectives and inductive biases differ.

Table 8. Single-run measured NDCG@20 / TR@20 / Coverage@20 (%). No uncertainty intervals are available; differences must not be interpreted as statistically significant.

Model	ML-100K	ML-1M	Gowalla	Yelp2018	Amazon-Book
CoopGCN	**0.1826 / 0.0106 / 46.9**	**0.1994 / 0.0075 / 40.7**	0.1089 / 0.0104 / 7.95	**0.0376 / 0.0006 / 5.36**	**0.0235 / 0.0036 / 6.47**
LightGCN++	0.1627 / 0.0100 / 46.5	0.2054 / 0.0008 / 37.3	**0.1092 / 0.0116 / 8.06**	**0.0359 / 0.0006 / 5.01**	0.0222 / 0.0027 / 5.44
GAT-CF	**0.1943 / 0.0022 / 19.4**	0.2096 / 0.0001 / 10.0	0.0038 / 0.0005 / 10.3	0.0014 / 0.0003 / 0.13	0.0002 / 0.0000 / 0.05
LightGCN	0.1842 / 0.0036 / 27.8	0.2128 / 0.0000 / 9.82	0.0348 / 0.0000 / 0.20	0.0145 / 0.0000 / 0.35	0.0002 / 0.0000 / 0.05
HPCF	0.1747 / 0.0020 / 25.7	**0.2141 / 0.0000 / 9.23**	0.0278 / 0.0000 / 0.39	0.0110 / 0.0000 / 0.48	0.0003 / 0.0000 / 0.05
HCCF	0.1723 / 0.0019 / 21.2	0.2070 / 0.0000 / 8.23	0.0291 / 0.0000 / 0.35	0.0113 / 0.0000 / 0.25	0.0002 / 0.0000 / 0.05
DyHuCoG	0.1718 / 0.0026 / 20.5	0.2114 / 0.0000 / 9.42	0.0338 / 0.0000 / 0.32	0.0144 / 0.0000 / 0.35	0.0002 / 0.0000 / 0.05
RecDCL	0.3103 / 0.0000 / 13.1	0.2997 / 0.0000 / 5.42	0.0434 / 0.0000 / 0.27	0.0109 / 0.0000 / 0.25	0.0042 / 0.0000 / 0.05
NCF	0.2939 / 0.0000 / 6.54	0.2883 / 0.0000 / 4.80	0.0444 / 0.0000 / 0.79	0.0129 / 0.0000 / 0.84	0.0062 / 0.0001 / 0.54
MF	0.1467 / 0.0052 / 34.5	0.2043 / 0.0003 / 17.1	0.0004 / 0.0006 / 6.70	0.0006 / 0.0006 / 6.82	0.0003 / 0.0000 / 0.05

7.2. RQ1: accuracy is regime-dependent

Within the direct hypergraph/cooperative peer group (HCCF, HPCF and DyHuCoG), CoopGCN has the highest point-estimate NDCG@20 on four of five datasets. It is not uniformly superior. On ML-1M its 0.1994 is below HPCF (0.2141), LightGCN (0.2128), DyHuCoG (0.2114), GAT-CF (0.2096), HCCF (0.2070) and LightGCN++ (0.2054). On Gowalla it is effectively tied with LightGCN++ (0.1089 versus 0.1092). On Yelp2018 and Amazon-Book it has the highest graph-model point estimate (0.0376 and 0.0235), only 4.7% and 5.9% above LightGCN++. Single-run evidence cannot establish that these small margins survive seed or tuning variation.

The non-graph context changes the dense-data interpretation. RecDCL reaches 0.3103 and 0.2997 on ML-100K and ML-1M, respectively, while NCF reaches 0.2939 and 0.2883—roughly 35–60% above the best graph-model point estimates. Both methods have zero TR@20 on these datasets and markedly lower coverage, so they occupy a head-accuracy extreme rather than uniformly dominating the measured outcomes. Nevertheless, their scale shows that graph-based credit assignment is not the strongest lever for dense-benchmark NDCG in this record. CoopGCN's case must therefore rest on a different accuracy–exposure operating point and on sparse-regime behavior, not on a general claim that graph propagation is the best dense-data architecture.

The retained ML-1M log has contrastive loss around 5.0 against ranking loss around 0.11, consistent with over-regularisation, but this post-hoc observation does not prove causation. A per-dataset λ_1 sweep and measured component ablation are needed.

7.3. RQ2: long-tail exposure and learned attention

The distributional point estimates are larger and consistent within the direct peer group. CoopGCN has the highest TR@20 and Coverage@20 among HCCF, HPCF and DyHuCoG on all five datasets. On ML-1M it reaches TR@20 =0.0075 and coverage =40.7%, while those three peers have TR@20 =0 and coverage between 8.23% and 9.42%. On Amazon-Book, its coverage is 6.47% versus 0.05% for each peer. LightGCN++ remains a strong comparator and slightly exceeds CoopGCN on both TR@20 and coverage on Gowalla.

GAT-CF is stronger in NDCG on ML-100K and ML-1M, but its NDCG falls to 0.0038, 0.0014 and 0.0002 on Gowalla, Yelp2018 and Amazon-Book. This could reflect a real sparse-regime limitation or an implementation/tuning failure: learned attention has greater capacity than fixed LightGCN weighting, and its roughly tenfold deficit to LightGCN on Yelp2018 is anomalous. Because the retained artefacts contain neither learning-rate sweeps nor training curves, we cannot rule out the latter. The fair conclusion is that the measured configuration of GAT-CF collapsed, not that attention generally fails. The same standard applies to MF: its NDCG@20 falls to 0.0004 on Gowalla and 0.0003 on Amazon-Book, far below its dense-data values, without retained tuning traces that distinguish a genuine sparse-regime failure from an implementation or optimisation artifact. Neither collapse supports an architecture-wide claim. The pre-specified three-part victory condition is only partly met: CoopGCN leads the retained GAT-CF configuration on tail recall and coverage, while measured comparative noise resilience is absent.

7.4. Descriptive accuracy–coverage sensitivity

A weaker ranker could mechanically expose more items, so coverage gains alone do not identify a credit-assignment effect. As a descriptive sensitivity check, Table 9 reports Spearman rank correlation between NDCG@20 and Coverage@20 across the ten model point estimates and, separately, across the seven graph models. The calculation is regenerated from data/measured\$_r$esults.csv by scripts/analyze\$_m$easured\$_t$radeoff.py.

Table 9. Descriptive Spearman between NDCG@20 and Coverage@20 across single-run model point estimates. This is a sensitivity diagnostic, not a seed-level test or causal estimate.

Dataset	All models (n=10)	Graph models (n=7)
ML-100K	-0.636	-0.179
ML-1M	-0.842	-0.607
Gowalla	0.006	-0.107
Yelp2018	0.305	0.775
Amazon-Book	0.841	0.885

The signs are not uniformly negative: the dense benchmarks show an accuracy–coverage trade-off, Gowalla is near zero, and the two largest sparse catalogues show positive association. This rules out a universal mechanical explanation in which lower NDCG necessarily produces higher coverage, but it does not isolate CoopGCN's mechanism. The sample is only 7–10 heterogeneous models per dataset, contains ties, and inherits all single-run and tuning limitations of Table 8; measured component ablations remain necessary for causal attribution.

7.5. RQ3–RQ4: component attribution and random-noise resilience remain open

The earlier component and random-edge-injection tables used projected full model values, including ML-1M NDCG@20 =0.2960, that conflict with the measured 0.1994. They cannot support claims that $\mathbf{G_1}$ accounts for 62.2% of tail recall or that CoopGCN loses only 5.4% under 20% injection. We therefore remove those tables and answer neither RQ3 nor RQ4 empirically. Required confirmatory experiments are measured leave-one-component-out runs, including w/o L_{game}, and independently retrained 0/5/10/20% random-injection curves for CoopGCN, GAT-CF, LightGCN++ and LightGCN. Proposition 2 remains a local Channel-A statement and must not be read as a substitute for these experiments.

7.6. RQ5: serving-path computation is reduced, not timed

Equation (9) removes Monte-Carlo Shapley estimation from the serving path by construction. It does not make inference identical to LightGCN: the learned attention network still executes once per edge. No wall-clock latency, training-time overhead, peak memory or distillation-gap measurement is retained. RQ5 therefore remains open.

7.7. Discussion

The measured evidence suggests an accuracy–exposure frontier rather than uniform dominance. CoopGCN changes which items are exposed, and the effect is largest against direct hypergraph/cooperative peers in sparse catalogues. On dense ML-1M, that redistribution coincides with lower NDCG. Because every cell is a single run and baseline tuning traces are incomplete, even this regime-dependent interpretation is a hypothesis supported by preliminary point estimates, not a statistically established result. Accordingly, this manuscript should be read as a methods paper with Stage-1-style preliminary validation: the architecture, local theory, leakage controls and falsification criteria are complete, whereas confirmatory component, noise, timing, attribution and multi-seed studies are not.

8. Conclusion, limitations and future work

8.1. Conclusion

CoopGCN proposes a tri-level, Shapley-derived credit-assignment framework for graph collaborative filtering. The exact credits satisfy the Shapley axioms; the deployed sigmoid- and attention-based weights do not. They retain only a credit-ordering signal within a fixed degree context, subject to Monte-Carlo and distillation error. Channel A exactly recovers LightGCN propagation at λ=0 when the other channels are disabled, and Proposition 2 bounds only a single edge's Channel-A deviation from its zero-credit reference.

Single-run measurements suggest that CoopGCN substantially broadens tail and catalogue exposure relative to direct hypergraph/cooperative peers and remains competitive on sparse datasets. They also show a clear failure on dense ML-1M, where CoopGCN has the lowest NDCG among the measured graph models. The data do not establish universal accuracy, robustness, explainability or efficiency superiority. The contribution at this stage is a coherent architecture, carefully delimited theory, and preliminary evidence for an accuracy–exposure trade-off that now requires confirmatory evaluation.

8.2. Limitations

Single-run evidence. No seed variance, confidence interval or significance test is available. Small sparse-data margins over LightGCN++ and the sign of the dense-data deficit require at least five independent runs or user-level bootstrap intervals before comparative claims are warranted.

Projection artefacts are not evidence. Earlier projected accuracy, component and random-noise tables are contradicted by measurement and have been removed from the results section. Consequently RQ3 and RQ4 remain open.

Baseline tuning cannot be audited completely. Shared splits and the evaluator are available, but trial-level logs and GAT-CF training curves are not. The sparse GAT-CF collapse may be a tuning or implementation artefact.

Theory is local. Proposition 2 is a per-edge, per-layer Channel-A bound relative to a non-zero reference. It neither bounds multi-layer/full-model perturbation nor NDCG degradation. The full weight retains degree normalisation, and efficiency and additivity apply to ϕ, not deployed weights.

Attribution is not validated. Credits are structurally available, but no deletion, insertion, stability or explainer-comparison experiment has been executed. They must not be described as faithful explanations.

Efficiency is not measured. Serving omits Shapley sampling but retains an attention network. Training time, inference latency, memory and the worst-case distillation gap δ are unknown.

Scope and validation reuse. Hyperedges depend on metadata; all metrics are reported at one cut-off; random corruption and targeted poisoning have not been measured in the retained evidence; and complete preprocessing manifests and hyperparameter-search traces are unavailable. Repeated $\mathbf{G_3}$ valuation on the same validation split may overfit that holdout; a nested valuation split is required in confirmatory work.

8.3. Future work

The immediate confirmatory study should (i) run at least five seeds with a fixed and published search budget; (ii) retain training curves and selected hyperparameters for GAT-CF and every comparator; (iii) execute measured component, w/o L_{game} and random-injection ablations; (iv) report wall-clock and memory measurements; and (v) execute the attribution protocol below. Until those experiments are complete, extending the model to targeted attacks, metadata-free graphs, sequential recommendation or streaming graphs is secondary.

8.4. A pre-specified protocol for evaluating the attributions

Because the explainability claim is the one most likely to be over-read, we fix in advance what would falsify it. Four tests, in increasing order of stringency.

Deletion (necessity). Remove the top-k interactions of user u by ϕ_{ui}, re-score without retraining, and record the drop in s_{ui} for the target item. Compare against removing k random interactions and k highest-degree interactions. A faithful attribution must produce a steeper degradation curve than both.

Insertion (sufficiency). Starting from the empty history, re-introduce interactions in descending ϕ_{ui} order and record how quickly s_{ui} recovers. Report the area under both curves, the standard comprehensiveness and sufficiency summary.

Stability. Recompute ϕ_{ui} across seeds and across $R \in \{4, 8, 16, 32\}$ permutations, and report rank correlation between runs. Monte-Carlo attribution with $R = 8$ may not be rank-stable; if it is not, the attributions are unusable as explanations even if the recommender itself performs well.

Comparison. Repeat all three against GNNExplainer [26], PGExplainer [27] and SubgraphX [28], the last being the closest competitor since it also uses Shapley values on graphs.

The victory condition is that ϕ dominates degree and random baselines on deletion and insertion *and* attains rank correlation above 0.8 across seeds. Until these are run, the explainability contribution of this work is a design property, not a result.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics and research integrity

This study uses five publicly released recommendation benchmarks (MovieLens-100K, MovieLens-1M, Gowalla, Yelp2018 and Amazon-Book) and recruits no human participants. The main table reports single-run measured records; it does not report uncertainty or statistical significance. Projection-only leaderboards, ablations and random-noise curves from earlier drafts are not presented as findings. The measured Shapley-versus-attention comparison is partial: clean-data metrics are available, comparative noise resilience and a complete GAT-CF tuning audit are not. Per-interaction credits are potentially sensitive behavioural artefacts and, absent the pre-specified faithfulness tests, are not represented as validated explanations.

Declaration of generative AI use

During the preparation of this work the authors used a large language model to assist with language editing and LaTeX formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The datasets analysed in this study are publicly available: MovieLens-100K and MovieLens-1M from GroupLens, and Gowalla, Yelp2018 and Amazon-Book from the benchmark sources documented in the repository. Source code and the retained result records are available at https://github.com/mouadlouhichi/CoopGCN.

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